# PEDIATRICS CASE PRESENTATION

**MODERATOR : DR CHETHANA, IGICH**

**PRESENTER : DR HARSHA B J, BMCRI**

# PATIENT PARTICULARS

NAME : Master Akash

AGE : 7 years 6 months

SCHOOLING : studying in 3rd standard

GENDER : Male

ADDRESS : Nelamangala, Bengaluru

DOA : 22nd August

DOE : 23rd August

INFORMANT : Child himself and his mother
Mrs Kavya, reliable

(studied till SSLC)

# CHIEF COMPLAINTS :

- Easy fatigability * 1 month
- Increased pallor * 1 week
- Palpitations * 1 week

# HISTORY OF PRESENTING ILLNESS

Child was apparently healthy 1 month back when he started experiencing easy fatigability which was insidious in onset and gradually progressive. Previously he was able to carry out activities like playing, going and coming back from a shop about 1km without any difficulty, later he started experiencing fatigue while playing with friends and going to shop. This easy fatigability has now progressed and the child experiences fatigue even while doing daily activities like taking bath. Fatigue relieves on taking rest.

Mother has also noticed that child has become pale since a week, which is generalised.

Child is also experiencing palpitations i.e awareness of heart beat since a week, sudden in onset and present while doing daily activities like taking bath, wearing clothes. It is regularly and aggravates on exertion and relieved on taking rest. There is no associated sweating or passage of urine after the event.

No h/o chest pain, breathlessness, difficulty in breathing on lying down or sitting erect and transient loss of consciousness (syncope), no h/o swelling of extremities (oedema), decreased urine output.

No h/o vomiting of blood (hematemesis), passage of black tarry stools (malena), passage of blood per rectum (haematochezia), passage of blood through nose (epistaxis) passage of blood in urine (haematuria), gum bleeding or purplish discoloration of skin.

No h/o blood transfusions.

No h/o fever with chills and rigors or rashes or night sweats.

No h/o abdominal pain, abdominal distension, no h/o diarrhoea, no h/o passage of worms in stools or vomitus.

No h/o loss of weight, pain in bones.

No h/o constipation, weight gain, cold intolerance.

No h/o exposure to radiation, drug usage which would cause bone marrow suppression or haemolysis.

No h/o yellowish discoloration of eyes and skin.

# PAST HISTORY :

No h/o similar complaints in the past i.e h/o easy fatigability, pallor or palpitations.

No h/o previous hospitalisation or surgeries.

No h/o blood transfusions in the past.

No history suggestive of any chronic infection.

No h/o tuberculosis, asthma, epilepsy.

# BIRTH HISTORY :

## ANTENATAL HISTORY

Registered case at Nelamangala government hospital, Bengaluru.

## FIRST TRIMESTER :

Pregnancy was confirmed after one month of missed periods at hospital by UPT.

History of spontaneous conception.

1st trimester scan (dating scan) was done and found to be normal.

Folic acid tablets were taken.

No h/o burning micturition, fever with rashes, excessive vomiting.

No h/o spotting or bleeding per vagina.

No h/o exposure to radiation and drug intake.

Regular ANC was done, weight gain was 1kg.

## SECOND TRIMESTER :

Quickening was felt at $5^{th}$ month.

2nd trimester scan (Anomaly scan) was done and found to be normal.

2 doses tetanus toxoid were taken.

Iron and calcium tablets taken, weight gain was 6 kg.

No h/o fever, increased frequency/burning of micturition.

No h/o pedal oedema, blurring of vision, headache.

## THIRD TRIMESTER :

Continued perception of foetal movements.

Iron and calcium tablets taken, weight gain was 5 kg.

Obstetric scan was done and found to be normal.

No h/o fever, increased frequency/burning of micturition.

No h/o pedal oedema, blurring of vision, headache.

No h/o of pain abdomen, leaking or bleeding per vagina.

# NATAL HISTORY

LMP : 1st June 2011 , EDD : 7th March 2012

DOB : 4th March 2012

Birth order : 1  Place of delivery : Nelamangala Government Hospital

Normal vaginal delivery at term and baby cried immediately after birth &  Breast feeding initiated within half an hour of delivery.

Birth weight of the baby : 3.1kg

# NEONATAL HISTORY

No h/o of NICU admission, umbilical catheterisation or exchange transfusion.

No history of neonatal jaundice, phototherapy.

Exclusively breastfed for 6months and continued for 1.5 years

Complementary food was given after 6 months.

# NUTRITIONAL HISTORY

| TIMINGS | Food item and amount consumed | Calorie intake (kcal) | Protein (g) |
|---|---|---|---|
| **MORNING** | 2 Idly + Sambar<br>1 Glass Milk | 300<br>110 | 5<br>3 |
| **AFTERNOON** | 1Chapati + 2 Bowl Rice + Sambar | 530 | 10 |
| **EVENING** | 1 Cup Tea | 80 | 2 |
| **NIGHT** | 1Chapati + 1bowl Rice + Sambar | 360 | 6 |
| **TOTAL** | | 1380 | 26 |
| **REQUIRED** | | 1900 | 32 |
| **DEFICIT** | | **520 (27%)** | **6 (18%)** |

Child consumes foods rich in iron and other factors necessary for haemoglobin synthesis such as green leafy vegetables, curd also meat occasionally.

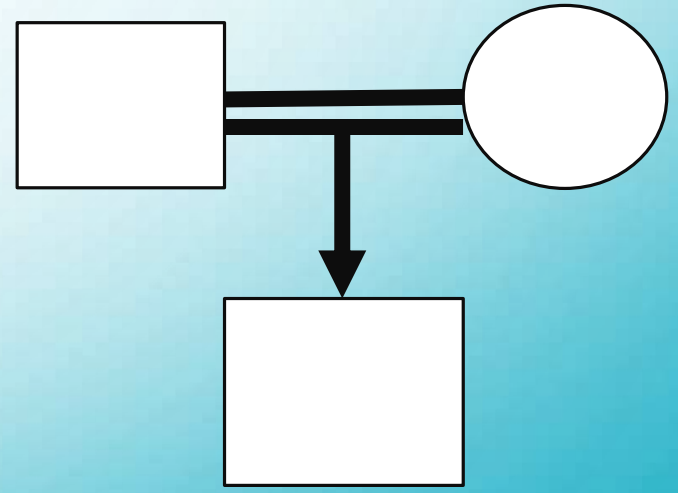
## DEVELOPMENTAL HISTORY

Developmental milestones have been achieved regularly and appropriate for age.

Good scholastic performance and there is no h/o decrease in this since one month.

## IMMUNIZATION HISTORY

Immunized up to date.



## FAMILY HISTORY : 3 members

2nd degree consanguineous marriage

Married life - 9 years

No history of bleeding tendency, anemia, jaundice in other family members.

No h/o congenital anomalies in the family.

No h/o DM, HTN, epilepsy or asthma in family members.

## PERSONAL HISTORY :

Diet - mixed, predominantly vegetarian

Appetite - normal

Sleep - sound

Bowel and bladder - normal and regular

Patient denied any having habits

## SOCIOECONOMIC HISTORY

Father is a teacher, graduate monthly income around ₹ 20000.

Belongs to **Upper Middle Class** according to modified Kuppuswamy classification

## ENVIRONMENTAL HISTORY

Lives in a Kutcha house without any overcrowding and with good ventilation.

Hand hygiene satisfactory, hygienic toilet practices, waste segregation practiced.

Usage of filtered water for drinking and cooking.

# SUMMARY :

A 7 year 6 months old male child who is first born to 2nd degree consanguineously married couple presented with progressive easy fatigability since 1 month, increased pallor and palpitations since 1 week. The child consumes a predominantly vegetarian diet which is 27% deficient in calories and 18% deficient in protein. He is immunized up to date with normal developmental milestones and belongs to upper middle class.

# GENERAL PHYSICAL EXAMINATION

The child is conscious cooperative and well oriented to time place and person, well built and poorly nourished.

Examined after taking consent from mother and in adequate light.

## VITAL SIGNS :

Pulse : 92 beats per minute, regular rhythm, good volume, normal character, no radio radial, no radio femoral delay.

Respiratory rate : 22 breaths per minute, abdominothoracic.

Blood pressure : 106/82mm Hg , measured in right brachial artery in sitting position - between $50^{th}$ to $90^{th}$ centile.

Temperature : 98°F , measured in axilla.

(vitals appropriate for age)

# HEAD TO TOE EXAMINATION

Hair - normal, no e/o thinning or loss of hair, loss of lusture.

Frontal bossing - present

Parietal bossing - present

Malar prominence - present

Eye - pallor is present, no icterus / micro cornea / subconjunctival haemorrhage.

Ears - normal

Nose - flat nasal bridge

Oral cavity - Good Hygiene, pallor is present, no e/o angular cheilitis or stomatitis

Tongue - normal no e/o stomatitis and other abnormalities

Dentition – normal, no e/o malocclusion.

Skin - pallor, no hyper pigmentation or petechiae.

Upper Limb – Normal no e/o bifid humerus, absent radius, triphalangeal thumb.

Palmar creases are seen.

Nails - normal no e/o platynychia / koilonychia

Lower Limb – Normal, no e/o any ulcers.

Chest
Spine
Back
} normal

Cyanosis
Clubbing
Lymphadenopathy
Oedema
} not seen

External genitals - normal

# ANTHROPOMETRY

| PARAMETERS | OBSERVED | EXPECTED | Centile |
|---|---|---|---|
| HEIGHT FOR AGE | 130cm | 123.5cm | $50^{th}$ - $90^{th}$ |
| WEIGHT FOR AGE | 18kg | 23.3kg | $10^{th}$ - $50^{th}$ |
| BMI | 11.7kg/m2 | 15.3kg/m2 | Below $3^{rd}$ |

**IMPRESSION :** The child is thin with low BMI indicative of acute malnutrition.

# SYSTEMIC EXAMINATION :

**PER ABDOMEN EXAMINATION**

**INSPECTION** :

Abdomen is normal in size and shape.

Umbilicus central and inverted.

Corresponding quadrants move with respiration.

No visible pulsations, no dilated veins, no scars.

Hernial orifices normal.

**PALPATION :**

**Superficial palpation :** Inspectory findings are confirmed

No local rise of temperature, no tenderness.

No guarding rigidity.

**Deep palpation** - A Mass in right hypochondrium, firm in consistency with smooth surface and sharp boarder extending 3cm below right costal margin. It moves with respiration. These findings are suggestive of liver.

A mass is also felt in left hypochondrium which is firm in consistency with smooth surface, sharp boarder and notched margins, it moves with respiration and extends 8cm below left costal margin towards umbilicus. These findings are suggestive of spleen.

**PERCUSSION** :

Dullness in the right hypochondrium extending 3cm below the coastal margin in the midclavicular line.

Dullness in the left hypochondrium extending 8cm along the line joining coastal margin to umbilicus.

**LIVER SPAN** : 12 cm ( increased for age and sex )

No evidence of free fluid.

**AUSCULTATION** : bowel sounds heard.

# EXAMINATION OF OTHER SYSTEMS :

**Cardiovascular System :**

**Inspection :**

Precordium appears normal

Apical beat - left 5th intercostal space ½" medial to midclavicular line

No visible pulsations, scars, sinuses or bony deformities.

**Palpation :**

Inspectory findings confirmed with apical impulse at left 5th intercostal space ½" medial to midclavicular line, normal in character

Auscultation : S1 and S2 heard in all areas.

Pulmonary area - High pitched soft midsystolic murmur of grade 2 best heard with diaphragm of the stethoscope without conduction and radiation.

**RS** - Normal vesicular breath sounds heard Bilateral equal air entry no added sounds

**CNS** - Higher mental function normal no sensory or motor deficit

# SUMMARY :

A 7 year 6 months old male child who is first born to 2nd degree consanguineously married couple presented with progressive easy fatigability since 1 month increased pallor and palpitations since 1 week. The child consumes a predominantly vegetarian diet which is 27% deficient in calories and 18% deficient in protein. General survey revealed that the child was found to be pale, with features of frontal and parietal bossing, malar prominence and flat nasal bridge, anthropometry revealed a BMI of less than 3$^{rd}$ centile, systemic examination revealed hepatomegaly with moderate splenomegaly and a Hemic (flow) murmur at pulmonary area.

# DIAGNOSIS

Haemolytic anaemia probably Thalassemia intermedia, not in failure with acute malnutrition.

Acute leukaemia.

Storage disorder - Gaucher’s disease.

I would like to do further investigations to confirm the diagnosis.

# THANK YOU